

# Duc-Cuong Vu

[vdcuong2002@gmail.com](mailto:vdcuong2002@gmail.com) <https://duccuongvu.github.io>

First-year PhD student at Nanyang Technological University (NTU), Singapore, researching nonlinear control and robotics, with a focus on safety-critical control, model predictive control, and dynamic robotic systems.

## Education

**Ph.D. in Electrical and Electronic Engineering**, Nanyang Technological University ([NTU](#)), Singapore Aug 2026 - present  
Funded by: NTU Research Scholarship

**Master of Sci. in Control Engineering and Automation**, Hanoi University of Science and Technology ([HUST](#)), Jul 2024 - Jul 2026  
**Thesis:** Design control structures for parallel platforms in maritime applications  
Funded by: Master, PhD Scholarship Programme of Vingroup Innovation Foundation ([VinIF](#))

**Bachelor of Sci. in Control Engineering and Automation**, Hanoi University of Science and Technology ([HUST](#)), Oct 2020 - Apr 2024  
Excellent degree, GPA: 3.71/4 (Rank: 22/499, top 5%). Finished the 4-year BSc program in just 3.5 years.

## Selected publications

- ★ Global trajectory generation and tracking control for AUVs with optimal coverage sensor networks  
*Ocean Engineering* (SCIE Q1) (2025) [[pdf](#)]  
**D.C. Vu**<sup>†</sup>, S. Tran<sup>†</sup>, T.L. Nguyen, and D.C. Hoang <sup>†</sup>: equally contribution
- ★ Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations  
*Ocean Engineering* (SCIE Q1) (2025) [[pdf](#)]  
**D.C. Vu**, D.H. Nguyen, M.N. Vu, and T.L. Nguyen
- ★ CBFs-based Model Predictive Control for Obstacle Avoidance with Tilt Angle Limitation for Ball-Balancing Robots  
*IEEE Access* (SCIE Q2) (2025) [[pdf](#)]  
M.D. Pham, **D.C. Vu**, T.T.H. Nguyen, T.V.A. Nguyen, M.N. Vu, and T.L. Nguyen
- ★ A novel approach of Consensus-based Finite-time DSMC for Stewart platform manipulators motion tracking  
*Results in Engineering* (ESCI Q1) (2025) [[pdf](#)]  
**D.C. Vu**, D.H. Nguyen, and T.L. Nguyen
- ★ Unifying Hierarchical Sliding Mode Control and Control Barrier Function for Tilt Angle Constraint of a Ball-Balancing Robot  
*IET Cyber-Systems and Robotics* (SCIE Q3) (2025) [[pdf](#)]  
T.T.H. Nguyen, **D.C. Vu**, M.D. Pham, T.V.A. Nguyen, and T.L. Nguyen
- ★ Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints  
*International Journal of Robust and Nonlinear Control - IJRNC* (SCIE Q1) (2024) [[pdf](#)]  
**D.C. Vu**, M.D. Pham, T.T.H. Nguyen, T.V.A. Nguyen, and T.L. Nguyen

## Other manuscripts

- ★ Decentralized Control Strategy with Consistent Joint-to-Actuator Control Law for Parallel Mechanisms on a Humanoid Platform  
Submitted manuscript [[pdf](#)]  
T.V. Trung, V.T. Nguyen, **D.C. Vu**, D.H. Nguyen, M.C. Nguyen, and M.N. Vu
- ★ Adaptive Whole-Body MPC with Optimal Posture Planning for Humanoid Robots under Unknown Payloads  
Submitted manuscript  
M.C. Nguyen<sup>†</sup>, **D.C. Vu**<sup>†</sup>, V.A. Vu, T. Nguyen, D.T. Pham, M.H. Phung, A.V. Ngo, and M.N. Vu <sup>†</sup>: equally contribution
- ★ Real-time Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators  
Submitted manuscript  
**D.C. Vu**, V.T. Nguyen, D.H. Nguyen, M.C. Nguyen, T.V. Trung, and M.N. Vu

## Experience

**R&D Robotics Engineer at VinRobotics, Vingroup** Sep 2025 - Jul 2026  
Full-time, supervised by [Dr. Minh Nhat Vu](#) Hanoi, Vietnam

- ★ Led dynamic parameter identification for robotic manipulators: system identification, kinematic calibration, and friction modeling.

- ★ Designed a real-time inverse kinematics solver achieving 100% success rate with execution time below 40  $\mu$ s.
- ★ Developed moDrdel-based control for dual-arm humanoid robots: fuzzy-based adaptive compliant control, computed torque control, gravity/friction compensation, and singular perturbation-based methods.
- ★ Implemented MPC-based motion planning for dual-arm coordination using receding horizon optimization with Acados and CasADi (C++/Python/MATLAB).
- ★ Designed time-optimal motion planning and trajectory generation, including Rubik's cube planning for single-arm manipulation.
- ★ Designed motion planning for humanoid standup/sitdown motions for object lifting.
- ★ Contributed to research activities, with papers submitted to *IROS 2026*, *IEEE-TASE*, and *IEEE RA-L*.

---

#### Research Project Assistant at HUST

Jan 2025 - Dec 2026

Contract (Hybrid), supervised by [PhD. Chinh Hoang Duc](#) and [Assoc.Prof.PhD. Tung Lam Nguyen](#)

Hanoi, Vietnam

- ★ Contributed to a project on autonomous robot navigation integrating sensor networks and wireless communication.
- ★ Developed a MuJoCo-based simulation framework for AUVs, incorporating underwater dynamics, sensing, disturbances, and communication constraints.
- ★ Implemented and validated control algorithms for navigation, obstacle avoidance, and trajectory tracking.
- ★ Collaborated on system integration and co-authored a peer-reviewed scientific publication.

---

#### Research Project Assistant at HUST

Mar 2025 - Dec 2025

Contract (Hybrid), supervised by [PhD. Minh Nhat Vu](#) and [Assoc.Prof.PhD. Tung Lam Nguyen](#)

Hanoi, Vietnam

- ★ Worked on advanced control of a ship-mounted Stewart platform for marine applications (funded by KIST).
- ★ Designed robust and safety-critical control strategies for parallel robotic systems under disturbances.
- ★ Conducted modeling, simulation, and validation for nonlinear dynamic systems.

---

#### Graduate Researcher at HUST

May 2024 - Jul 2026

Part-time (Hybrid), supervised by [Assoc.Prof.PhD. Tung Lam Nguyen](#)

Hanoi, Vietnam

- ★ Conducted research on control of parallel robotic systems for maritime applications.
- ★ Designed and implemented advanced control algorithms for a Stewart platform, validated through high-fidelity simulations (Simscape, MuJoCo).
- ★ Built experimental setups including mechanical assembly, hardware integration, Linux real-time systems, and EtherCAT communication.
- ★ Collaborated on publications and presented results to academic and industrial partners.

---

#### Student Intern at HUST

Oct 2021 - Apr 2024

Internship (Hybrid), supervised by [Assoc.Prof.PhD. Tung Lam Nguyen](#)

Hanoi, Vietnam

- ★ Developed mathematical models and simulation environments for 3D ballbot systems, focusing on nonlinear dynamics and trajectory generation.
- ★ Investigated advanced control strategies and implemented them in simulation and experimental setups.
- ★ Authored and co-authored peer-reviewed journal papers, including publications in *International Journal of Robust and Nonlinear Control* and *IEEE Access*.

---

## Academic activities

#### Invited review for

- ★ *Nonlinear Dynamics*
- ★ *International Conference on Intelligent Robots and Systems (IROS, 2026)*
- ★ *IEEE Robotics and Automation Letters (RA-L)*
- ★ *IEEE RO-MAN2026*
- ★ *Humanoid 2026*

---

#### Teaching, Seminars and Talks

2025: Talk "MuJoCo for Advanced Physics Simulation: From manipulators to autonomous vehicles" for "Motion Control" master course at HUST and MoCAR seminar [[pdf](#)]

2025: Seminar "Underwater Vehicles" for modeling training of Autonomous Underwater Vehicle at MEG-MoCAR [[pdf](#)]

---

## Honours & awards

NTU Research Scholarship    Nanyang Technological University (NTU), Singapore

---

Master, PhD Scholarship Programme    Vingroup Innovation Foundation (VINIF)

**Skills**

---

|                                     |                                                                                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>                  | Python, C/C++, MATLAB                                                                                                                   |
| <b>Robotics &amp; Control</b>       | Nonlinear control, Model Predictive Control (MPC), Control Barrier Functions (CBF), motion planning, system modeling and identification |
| <b>Tools &amp; Frameworks</b>       | ROS2, MuJoCo, Simulink/Simscape, Pinocchio, CasADi, Acados, VAMP, CAPT, Git                                                             |
| <b>Systems &amp; Implementation</b> | Real-time Linux, IgH EtherCAT, hardware integration, motion capture systems                                                             |
| <b>Research</b>                     | Scientific writing, peer-reviewed publications, technical presentations, literature review, experimental validation                     |